

MLDM Assignment 5

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 5

This are your test results

Knowledge Questions

☰ Expectation Maximization

Status

Seen but not answered

Your score

0 / 6

0%

Response

Answer with *Right* or *Wrong* the following statements.

Notes:

- Correct answer: +1 pts
- Wrong answer: -0.5 pts
- Unanswered: 0 pts

Unanswered	Right	Wrong	
☒	☐	☐	Expectation-Maximization (EM) is a generalization of K-Means
☒	☐	☐	Expectation-Maximization (EM) is a statistical approach
☒	☐	☐	The assumption behind the Expectation-Maximization (EM) algorithm is that the data can be explained by a mixture of parametrized probability distributions
☒	☐	☐	While working on a clustering problem, the true distribution of the data is always known.
☒	☐	☐	It is common to use K-Means to find the initial clusters before performing the Expectation-Maximization (EM).
☒	☐	☐	The Expectation-Maximization (EM) algorithm is an iterative approach that cycles between three steps: (Expectation step, Maximization step, and Minimization).

► Solution

☰ Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

Status Not started

Your score 0 / 6 0%

Response

Answer with *Right* or *Wrong* the following statements.

Notes:

- Correct answer: +1 pts
- Wrong answer: -0.5 pts
- Unanswered: 0 pts

Unanswered	Right	Wrong	
☐	☐	☐	DBSCAN was proposed to solve the problem of huge time and space requirements when handling large datasets.
☐	☐	☐	DBSCAN is not good at discovering clusters with arbitrary shapes (for example non-convex)
☐	☐	☐	The logic behind using DBSCAN is that the density of data points inside a cluster is much higher than the density outside the cluster.
☐	☐	☐	Epsilon-neighborhood is a square of width epsilon drawn around a point p to scan the number of points falling inside the square.
☐	☐	☐	In DBSCAN if a point does not satisfy the Minpts in the epsilon-neighborhood condition, the point is then marked as noise.
☐	☐	☐	DBSCAN requires a lot of domain knowledge in order to perform the algorithm

► Solution

☰ Minimum Spanning Tree Clustering (MSTC)

Status Not started

Your score 0 / 6 0%

Response

Answer with *Right* or *Wrong* the following statements.

Notes:

- Correct answer: +1.5 pts
- Wrong answer: -0.5 pts
- Unanswered: 0 pts

Unanswered	Right	Wrong	
☐	☐	☐	The MSTC algorithm represents data as a graph.
☐	☐	☐	In MSTC the distances are weighted edges.
☐	☐	☐	In MSTC we have only one parameter k defined as "neighborhood" of each edge.
☐	☐	☐	MSTC and Agglomerative Hierarchical Clustering (AHC) are very similar in terms of how they work.

► Solution

≡ Essay

Status

Not started

Your score

0 / 15

0%

Response

In your own words, explain how Agglomerative Hierarchical Clustering (AHC) works. **(including formulas is not mandatory)**

0 word

► Solution

Expectation maximization

📊 Expectation 1

Status

Not started

Your score

0 / 9

0%

Response

Given the following dataset that contains the incomes and level of education of 9 employees:

Income	Education
157434	0
261952	2
144657	0
118777	1
147511	1
163025	2
120131	0
162745	0
155569	1

Our task is to cluster people by their income, there are 3 clusters (Education = 0, 1, or 2).

Assume we are performing the EM algorithm and currently, we are in iteration **2**, such that:

• The distribution of all clusters is equal to: $f_n(x) = \frac{1}{\sigma\sqrt{2\pi}}exp\frac{-(x-\mu)^2}{2\sigma^2}$

• $\mu_1 = 188014.34, \sigma_1 = 52541.39$

• $\mu_2 = 143104.34, \sigma_2 = 18330.95$

• $\mu_3 = 146148.34, \sigma_3 = 18628.83$

• **The variance in this case is the population variance.**

Calculate the following:

• $f_{\mu_1, \sigma_1}(162745) =$

• $f_{\mu_2, \sigma_2}(162745) =$

• $f_{\mu_3, \sigma_3}(162745) =$

Calculate the following weights (index 6 = the income 163025):

• $z_{6,1} =$

• $z_{6,2} =$

• $z_{6,3} =$

► Solution

https://olat.vcrp.de/auth/RepositoryEntry/3909322134/CourseNode/106528579299712/test/0

3/7

Expectation 2

Status

Not started

Your score

0 / 9

0%

Response

Given the following dataset that contains the incomes and level of education of 9 employees:

Income	Education
157434	0
261952	2
144657	0
118777	1
147511	1
163025	2
120131	0
162745	0
155569	1

Our task is to cluster people by their income, there are 3 clusters (Education = 0, 1, or 2).

Assume we are performing the EM algorithm and currently, we are in iteration **2**, such that:

- The distribution of all clusters is equal to: $f_n(x) = \frac{1}{\sigma\sqrt{2\pi}}exp^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- $\mu_1 = 188014.34, \sigma_1 = 52541.39$
- $\mu_2 = 143104.34, \sigma_2 = 18330.95$
- $\mu_3 = 146148.34, \sigma_3 = 18628.83$
- The variance considered in this case is the population variance.

Calculate the following for income = 144657

- $f_{\mu_1,\sigma_1}(144657) =$
- $f_{\mu_2,\sigma_2}(144657) =$
- $f_{\mu_3,\sigma_3}(144657) =$

Calculate the following weights (index 3 = the income 144657):

- $z_{3,1} =$
- $z_{3,2} =$
- $z_{3,3} =$

► Solution

Maximisation

Status

Not started

Your score

0 / 18

0%

Response

Given the following dataset that contains the incomes and level of education of 9 employees: **(This task is the completion of the previous one)**

Income	Education
157434	0
261952	2
144657	0
118777	1
147511	1
163025	2
120131	0
162745	0
155569	1

Our task is to cluster people by their income, there are 3 clusters (Education = 0, 1, or 2).

Assume we are performing the EM algorithm and currently, we are in iteration **2**, such that:

- The distribution of all clusters is equal to: $f_n(x) = \frac{1}{\sigma\sqrt{2\pi}}exp^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- $\mu_1 = 188014.34, \sigma_1 = 52541.39$
- $\mu_2 = 143104.34, \sigma_2 = 18330.95$
- $\mu_3 = 146148.34, \sigma_3 = 18628.83$
- The variance considered in this case is the population variance.**

Calculate the weights for all data points and update the new model parameters:

$\mu_1 =$

$\sigma_1 =$

$\mu_2 =$

$\sigma_2 =$

$\mu_3 =$

$\sigma_3 =$

► Solution

Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

For this section you are given the following dataset: Country-data.csv, your task is to perform the DBSCAN clustering algorithm on this dataset.

DBSCAN Algorithm

Status

Not started

Your score

0 / 20

0%

Response

Write a simple DBSCAN algorithm that runs on the given dataset. **You are free to use any library of your choice**

Note:

- It is recommended to use scikit-learn
- Epsilon-neighborhood equals 2
- minPts equals 5

0 word

► Solution

Clusters

Status

Not started

Your score

0 / 10

0%

Response

According to your algorithm, to which clusters do the following countries belong?

USA

Cameroon

Germany

China

Australia

Cluster 1

Cluster 2

Cluster 3

Cluster 4

Cluster 5

Cluster 6

► Solution

 Upload file

Status	Not started		
Your score	0 / 1		0%

Response

Upload the Jupyter notebook containing the DBSCAN solution.

Note:

- The upload is mandatory in order to get the score for the programming task.

File

No file uploaded

Solution